MILESTONE 2

Data Understanding and Pipeline Setup

Student Guide and Grading Rubric

Module	Data Handling and Infrastructure for AI
Deadline	Sunday, 1st November 2026 at 23:59
Submission	Moodle: one PDF report and a link to the project repository
Assessment focus	Initial report on the data and setup of a reproducible data pipeline
Weightage	10% of the overall module grade

Purpose. *Milestone 2 demonstrates that you understand your selected data and can move it through a documented, repeatable pipeline. A final model is not required at this stage.*

Carry forward, do not duplicate	You may use Milestone 1 as the foundation for this report. If the project definition, data-split strategy or storage decision is unchanged, summarise or reference it briefly. Marks in Milestone 2 are based on new evidence: EDA, implementation, validation, integration and justified changes.

What you should have completed

- A confirmed project question and a clearly defined analytical, prediction, classification or forecasting task.
- A justified dataset, including its origin, licence, structure, limitations and ethical implications.
- Purposeful EDA that is interpreted in your own words and informs data-quality, preprocessing, splitting and pipeline decisions.
- A reproducible process for acquiring, storing, validating, cleaning and transforming the data.
- A documented repository that another student could set up and run from the instructions provided.

Learning outcomes addressed

This milestone provides evidence for data comparison and feature engineering, data-pipeline design and implementation, and evaluation of suitable data-storage approaches.

1. Required submission

1.1 Milestone report

Submit one self-contained PDF report, excluding references and appendices. The report may build on Milestone 1, but unchanged material should not be reproduced at length. Use clear headings and include the following sections:

Section	Contents
Project definition	Current title, problem statement, intended users or setting, and the exact analytical or ML task. Explain any change from Milestone 1.

Data source	Source URL or DOI, data owner, licence or terms of use, collection context, permissions, privacy, bias and ethical considerations.
Data description	Unit of analysis, number of records, fields, data types, target variable where applicable, time range, class balance, missingness and known limitations.
Exploratory data analysis	Central focus of Milestone 2. Include relevant statistics and purposeful visualisations. Interpret every result and explain how it affects data quality, cleaning, transformation, feature engineering, splitting or evaluation.
Design/ Architecture	A diagram showing acquisition, ingestion, validation, storage, processing and output. State what is automated and what remains manual.
Implementation	Evidence that the pipeline runs on a representative sample or full dataset. Include commands, screenshots or concise outputs where useful. If storage or splitting was implemented early, demonstrate its integration and validation rather than repeating its original justification.
Next steps (Optional)	Unresolved data issues, intended baseline model and a plan for Milestone 3.

1.2 Repository evidence

- README.md with the project purpose, data source, environment setup and exact run instructions.
- A requirements file or environment specification with pinned or clearly stated dependencies.
- Scripts or notebooks for data acquisition, validation, cleaning, transformation and EDA.
- A clear folder structure. Do not commit credentials, API keys, personal data or unnecessarily large raw datasets.
- Meaningful commits showing development over time. The submitted link must be accessible to the lecturing team.

1.3 Pipeline diagram (Optional)

Include one readable architecture or data-flow diagram. At minimum, show:



The diagram must correspond to your implemented code and storage choices. A generic cloud diagram without a connection to your project will not receive full credit.

2. Technical expectations

- **Data acquisition:** Data can be downloaded, queried or loaded using a documented command, script or notebook. If access is manual, explain why and preserve a reproducible local entry point.
- **Validation:** Check schema, data types, required fields, duplicates, valid ranges and obvious inconsistencies. Failures should be visible rather than silently ignored.
- **Cleaning and transformation:** Implement justified handling of missing values, duplicates, categorical variables, scaling or normalisation, text/image preparation, dates or outliers as relevant.
- **Data splitting:** If the strategy is unchanged from Milestone 1, summarise it briefly. Show how it is implemented or validated and explain any changes. Use chronological, grouped or stratified splitting where appropriate.
- **Storage:** If the choice is unchanged from Milestone 1, summarise or reference it. Provide evidence that storage is implemented, integrated with the pipeline, and appropriate to scale, format and access pattern. Cloud storage is not mandatory for its own sake.
- **Reproducibility:** A clean environment should be able to run the documented pipeline without hidden steps, private file paths or embedded secrets.
- **Evidence quality:** Figures and tables must be legible, labelled and interpreted. Large raw notebook dumps do not substitute for analysis.

3. Grading rubric

Milestone 2 contributes 10% of the overall module grade. The rubric records performance out of 100 for clear feedback and is then scaled to the 10% module weighting.

Criterion	Wt.	Excellent 70–100%	Good 60–69%	Developing 40–59%	Insufficient 0–39%
Project definition and development	5	Precise task, users and success criteria; changes from Milestone 1 are justified.	Task is clear with minor omissions.	Task is broadly stated but lacks precision or justification.	Unclear, inconsistent or absent project definition.
Data provenance, governance and ethics	15	Ownership, licence, collection context, privacy, bias and limitations are critically evaluated.	Source and licence are clear; main governance and ethical issues are discussed.	Source is identified but provenance, licence or risks are treated superficially.	Source/permission is unclear, or material ethical and legal risks are ignored.
Data understanding and EDA	20	Accurate, purposeful analysis of structure, quality and distributions; visualisations are interpreted and drive decisions.	Sound EDA covers the main quality and distribution issues with reasonable interpretation.	Basic summaries are present but incomplete, weakly interpreted or disconnected from decisions.	EDA is absent, incorrect, largely automated without interpretation, or not relevant to the task.
Pipeline architecture and storage design	15	Coherent end-to-end design; storage and processing choices are technically justified and match the project constraints.	Clear pipeline with mostly appropriate choices and some justification.	Partial or generic design; important stages or justifications are missing.	No credible pipeline design or inappropriate, unexplained architecture.
Pipeline implementation	25	Working, modular pipeline with acquisition, validation and transformation; handles failures and produces repeatable outputs.	Core pipeline runs and implements the main stages; some robustness or modularity is missing.	Partially working pipeline or notebook with substantial manual steps and limited validation.	Little executable evidence, major failures, or code does not correspond to the claimed process.
Reproducibility, repository and documentation	10	Another user can set up and run the work; dependencies, structure, commits and instructions are clear; secrets/data are managed correctly.	Repository is organised and mostly reproducible with minor documentation gaps.	Repository is accessible but setup or execution requires unstated steps.	Repository is inaccessible, disorganised, unsafe or not reproducible.
Next-step plan	10	Limitations and proposed baseline, evaluation approach and Milestone 3 plan.	Main limitations and next steps are identified and feasible.	General reflection with limited prioritisation or technical detail.	No credible reflection or plan for subsequent work.

Mark calculation: Each criterion receives a percentage judgement within the descriptor band. The weighted criterion marks produce a Milestone 2 score out of 100. This score is multiplied by 0.10 to calculate the contribution to the overall module grade. For example, 72/100 contributes 7.2 marks out of the available 10.

4. Submission and assessment conditions

How to submit

1. Upload the report as a single PDF to the Milestone 2 to Moodle.
2. Place the repository URL on the first page of the report and in the Moodle submission text.
3. Check repository access before submitting.
4. The Moodle timestamp determines whether a submission is on time. Keep a local copy of the submitted files.

Academic integrity and use of AI tools

You may use AI-assisted tools for learning, debugging, editing and code suggestions where permitted by the module policy. You remain responsible for every submitted claim, line of code, citation and result. Your submission must contain your own analysis and demonstrate that you understand the implementation.

- Acknowledge material use of generative AI, including the tool, purpose and how the output was checked or changed.
- Do not submit fabricated citations, datasets, results or implementation evidence.
- Do not upload confidential, personal, licensed or restricted data to an AI service without permission.
- Be prepared to explain and modify your code during feedback or demonstration.

Late submissions and exceptional circumstances

The standard SETU and module policies apply. If circumstances may affect your submission, contact the teaching team as early as possible and use the appropriate institutional process. Do not rely on a Slack message as a formal extension request.

Feedback

Feedback will identify the strongest evidence, priority corrections to the data or pipeline, and the work required before model training in Milestone 3. Slack discussion can support formative feedback but does not replace the submitted evidence.

Final clarification	
Carry forward, do not duplicate	• You may use Milestone 1 as the foundation for this report. If the project definition, data-split strategy or storage decision is unchanged, summarise or reference it briefly. • Marks in Milestone 2 are based on new evidence: EDA, implementation, validation, integration and justified changes.